

BADG 2: A Perfect Game

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Abstract

The earlier (b. a. d. g.) paper focused on a near-optimal strategy for maximizing expected value in gameplay. This paper focuses on a different question: what is the probability of playing a perfect game? Using standard recursive calculation, we calculate the probabilities with the out-of-box game setup. Then we venture into a related: What's the expected win probability with n standard dice of side count s as $n \rightarrow \infty$? We give poor bounds and prove convergence.

1 The Commercial Game

This is solvable using standard recursive techniques.

$$\vec{S} = [12, 1, 1, 1] \tag{1}$$

$$\vec{p} = [\frac{1}{6}, \frac{1}{8}, \frac{1}{10}, \frac{1}{12}] \tag{2}$$

$$P_{single}(n, k, \vec{p}) := \binom{n}{k} p^k (1-p)^{n-k} \tag{3}$$

$$P_{transition}(\vec{s}_1, \vec{s}_2, \vec{p}) := \prod_{i=0}^{dim(\vec{s})} P_{single}(\vec{s}_{1i}, \vec{s}_{2i}, \vec{p}_i) \tag{4}$$

$$F(\vec{0}) := 1 \tag{5}$$

$$substates(\vec{s}) := \times_{i=0}^{dim(s)} [0, s_i] \tag{6}$$

$$states = topological_sort(substates(\vec{S}) - [0, 0, 0, 0]) \tag{7}$$

$$for \vec{s} \in states : \tag{8}$$

$$F(\vec{s}) = \sum_{\vec{t} \in (substates(\vec{s}) - \vec{s})} P_{transition}(\vec{s}, \vec{t}) \cdot F(\vec{t}) \tag{9}$$

EV with $(d_{10}, d_{12}) = (0, 0)$	$d_6 \backslash d_8$	0	1
	0	1	0.125
	1	0.1667	0.05642
	2	0.07407	0.03244
	3	0.04192	0.02206
	4	0.02809	0.01679
	5	0.02107	0.01378
	6	0.01708	0.01195
	7	0.01463	0.01076
	8	0.01303	0.009966
	9	0.01195	0.009423
	10	0.01119	0.009044
	11	0.01065	0.008779
	12	0.01025	0.008591

EV with $(d_{10}, d_{12}) = (1, 0)$	$d_6 \backslash d_8$	0	1
	0	0.1	0.03469
	1	0.04556	0.02046
	2	0.02644	0.01425
	3	0.01815	0.01109
	4	0.01392	0.009299
	5	0.01152	0.008218
	6	0.01006	0.007537
	7	0.009124	0.0071
	8	0.008506	0.006818
	9	0.00809	0.006638
	10	0.007809	0.006529
	11	0.007618	0.006468
	12	0.007491	0.006441

$d_6 \backslash d_8$	0	1
0	0.08333	0.02908
1	0.03819	0.01726
2	0.02231	0.01209
3	0.0154	0.009464
4	0.01188	0.00798
5	0.009888	0.007088
6	0.008677	0.006532
7	0.007907	0.006181
8	0.007404	0.00596
9	0.007071	0.005825
10	0.006851	0.00575
11	0.006707	0.005715
12	0.006618	0.005709

EV with $(d_{10}, d_{12}) = (0, 1)$

$d_6 \backslash d_8$	0	1
0	0.02347	0.01087
1	0.01406	0.007809
2	0.009946	0.006252
3	0.007849	0.005385
4	0.006672	0.004879
5	0.005971	0.00458
6	0.005542	0.004408
7	0.005278	0.004318
8	0.00512	0.004283
9	0.005033	0.004284
10	0.004994	0.004312
11	0.004988	0.004356
12	0.005005	0.004413

EV with $(d_{10}, d_{12}) = (1, 1)$

Theorem 1.1. *The expected chance of a perfect game with r standard dice of s sides each, given $p := \frac{1}{s}$, is $F(r) = \sum_{k=1}^r \binom{r}{k} p^k q^{r-k} F(r-k)$, with $F(0)$ defined as 1.*

Proof: This is a clear application of the binomial theorem.

Bounding this quantity will be the principal thrust of this paper.

Lemma 1.2 (Easy Cases Lemma). *If $p = 0$, $F(r) = 0$. If $p = 1$, $F(r) = 1$. If $s = 2$, for any $r > 0$, $F(r) = \frac{1}{2}$.*

Proof: Though we won't be flipping infinite ($p = 0$) or 1-sided ($p = 1$), dice, these algebraic identities are obvious (can't succeed, can't fail, respectively) and will be useful later.

Base case for s=2: $F(1) = \frac{1}{2}$ since a perfect game occurs if and only if the coin flips 2 instead of 1.

Inductive case for s=2: Assuming $F(k) = \frac{1}{2}$ for all $0 < k < r$, and $F(0) = 1$:

$$\sum_{k=1}^{r-1} \binom{r}{k} p^k q^{r-k} F(r-k) + \binom{r}{r} p^r q^0 F(0) = \quad (10)$$

$$(2^r - 2) \frac{1}{2^r} \left(\frac{1}{2}\right) + \frac{1}{2^r} 2 \left(\frac{1}{2}\right) = \quad (11)$$

$$(2^r) \frac{1}{2^r} \left(\frac{1}{2}\right) = \frac{1}{2} \quad (12)$$

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- Wrong attempt: $F(t) < F(t-1)$?
- $F(n)$ exhibits provably asymptotic behavior

Theorem 1.3. *For an unbounded set of $r > 0$ standard s -sided dice, $s \geq 2$, the probability $F(r)$ of getting a perfect game has a lower bound of $\exp(-\frac{q}{(1-q)^2})$, if we define $q := 1 - \frac{1}{s}$.*

Proof:

- First, while at stage k , where k dice remain, the probability $P_s(k)$ of continuing to some stage $j < k$ is clearly $1 - q^k$.
- The probability $F(k)$ of passing all such stages is greater than or equal to $P^* = \prod_{i=1}^k P_s(k)$, since a successful traversal may involve skipping one or more stages, and P^* assumes we no stages.
- Therefore, $F(k) \geq \prod_{i=1}^k P_s(k) = \prod_{i=1}^k (1 - q^k)$.
- This means $\log(F(k)) \geq \sum_{i=1}^k \log(P_s(k)) = \sum_{i=1}^k \log(1 - q^k)$.

This $F(k)$ is apparently defined as the **q-Pochhammer symbol** $(q; q)_\infty = \prod_{i=1}^k (1 - q^i)$, with known convergence properties. I'll instead attempt to tackle convergence without the aid of these theorems.

Often, $\log(1 - z)$ is approximated by $-z$ near zero, but we are most concerned with z near one, since $1 - z$ approaches 0, and $\log(1 - z)$ explodes. Therefore, let's look at the expansion of $\log(1 - z)$:

- Identity 1: $f(z) = -\frac{1}{1-z} = -1 - z - z^2 - \dots$
- Identity 2: $\int f(z) = \log(1 - z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \dots$

So when summing $\sum_{i=1}^k \log(1 - q^i)$, we can instead sum

$$\log(F(k)) = \log(1 - q^1) + \log(1 - q^2) + \log(1 - q^3) + \dots \quad (13)$$

$$\log(F(k)) = (-q - \frac{q^2}{2} - \frac{q^3}{3} + \dots) + (-q^2 - \frac{q^4}{2} - \frac{q^6}{3} + \dots) + (-q^3 - \frac{q^6}{2} - \frac{q^9}{3} + \dots) + \dots \quad (14)$$

$$> (-q - q^2 - q^3 + \dots) + (-q^2 - q^4 - q^6 + \dots) + (-q^3 - q^6 - q^9 + \dots) + \dots \quad (15)$$

$$= \frac{-q}{1 - q} + \frac{-q^2}{1 - q^2} + \frac{-q^3}{1 - q^3} + \dots \quad (16)$$

$$> \frac{-q}{1 - q} + \frac{-q^2}{1 - q} + \frac{-q^3}{1 - q} + \dots \quad (17)$$

$$= \frac{1}{1 - q} [-q - q^2 - q^3 - \dots] \quad (18)$$

$$= \frac{-q}{(1 - q)^2} \quad (19)$$

- (2) follows from (1) by substitution using Identity1 above.
- (3) follows from (2) since the denominators of (2) are larger.
- (4) follows from (3) when dividing each sequence by $-q^k$ and applying identity 2.
- (5) follows from (4) since the denominators of (4) are larger.
- (6) and (7) follow from (5) by dividing out $\frac{1}{1-q}$ and applying Identity 1.

Since $\log(F(k)) > \frac{-q}{(1-q)^2}$, $F(k) > \exp(\frac{-q}{(1-q)^2})$.

s	p	q	bound
2	$\frac{1}{2}$	$\frac{1}{2}$	$\exp(-2) \approx .135$
3	$\frac{1}{3}$	$\frac{2}{3}$	$\exp(-6) \approx .00248$
4	$\frac{1}{4}$	$\frac{3}{4}$	$\exp(-12) \approx 6.14 \times 10^{-6}$
10	$\frac{1}{10}$	$\frac{9}{10}$	$\exp(-90) \approx 8.19 \times 10^{-40}$

Table 1: Very loose lower bounds on $F(k = \infty)$

This is a thoroughly unsatisfying lower bound, as we know $F(2) = \frac{1}{2}$.

Definition 1.4 ($(q; q)_\infty$: The q-pochhammer symbol). *The q-pochhammer symbol $(q; q)_\infty = \prod_{k=1}^\infty (1 - q^k)$ is a known convergent quantity.*

q	$(q; q)_\infty$	$\exp\left(-\frac{q}{(1-q)^2}\right)$
1/2	2.8879×10^{-1}	1.3534×10^{-1}
2/3	6.9272×10^{-2}	2.4788×10^{-3}
3/4	1.5545×10^{-2}	6.1442×10^{-6}
4/5	3.3680×10^{-3}	2.0612×10^{-9}
5/6	7.1400×10^{-4}	9.3576×10^{-14}
6/7	1.4913×10^{-4}	5.7495×10^{-19}
7/8	3.0815×10^{-5}	4.7809×10^{-25}
8/9	6.3155×10^{-6}	5.3802×10^{-32}
9/10	1.2861×10^{-6}	8.1940×10^{-40}

However, this does provide a positive lower bound, and brings up this interesting fact: **Though increasing the number of sides of the dice can drive $F(k)$ to zero, increasing k to ∞ will never set $F(k)$ to zero.**

If we can couple this with the proof of the next theorem, we can state that $F(k)$ has an asymptotic limit as $k \rightarrow \infty$ given some q .

Calculating by hand:

- $F(0) = 1$
- $F(1) = p$
- $F(2) = p^2(1 + 2q) = p^2(3 - 2p)$
- $F(3) = p^3(13 - 27p + 21p^2 - 6p^3)$
- $F(4) = p^4(75 - 316p + 606p^2 - 664p^3 + 432p^4 - 156p^5 + 24p^6)$

- $F(1) - F(0) = p - 1 = -(1 - p)$
- $F(2) - F(1) = p^2(3 - 2p) - p = (-1)p(1 - 2p)(1 - p)$
- $F(3) - F(2) = (-1)3p^2(1 - p)^3(1 - 2p)$
- $F(4) - F(3) = -p^3(1 - 2p)(p - 1)^2(12p^4 - 48p^3 + 72p^2 - 50p + 13)$

Theorem 1.5 (Incorrect Blind Alley: $F(n) < F(n - 1)$). *If $p < \frac{1}{2}$, $F(n - 1) > F(n)$ for $n > 1$.*

- TODO: Show calculation of various functions directly.
- Show calculation of $p = \frac{1}{2} - \epsilon$, $q = \frac{1}{2} - \epsilon$ with $p = 0.495$.
- Show interesting observations: growth of polynomial
- $R(t)$ is starting at position t , throwing $t - 1$ dice.
- $F(t) = p(1 + q^{t-1})F(t - 1) + qR(t) = pF(t - 1) + pq^{t-1}F(t - 1) + qR(t)$
- If $qR(t) < qF(t - 1) - pq^{t-1}F(t - 1)$, same as $F(t - 1) - R(t) > pq^{t-2}F(t - 1)$, then $F(t) < F(t - 1)$.
- In the special case $q = p = \frac{1}{2}$, then $F(t - 1) - R(t) = p^{t-1}(1) - p^{t-1} \cdot p + \text{other zero factors} = p^t = pq^{t-2} \cdot \frac{1}{2}$.
- So we're satisfied since $F(t) = \frac{1}{2}$ for $t \geq 1$.
- If $p \leq \frac{1}{3}$, $F(t - 1) - R(t) = [p^{t-1}F(0) - p^{t-1}F(1) + \text{other positive factors}] = p^{t-1}q + \epsilon$.

$$t = 3, F(3) - R(4) > pq^{4-1}F(4 - 1) \quad (20)$$

$$p^3F(0) - p^3F(1) \quad (21)$$

$$\binom{3}{2}p^2qF(1) - \binom{3}{2}p^2qF(2) \quad (22)$$

$$\binom{3}{1}p^2qF(2) - \binom{3}{2}p^2qF(3) \quad (23)$$

$$(24)$$